



David Sinden PhD

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British Citizen with permanent residency in Germany

Innovative applied mathematician with a strong track record in scientific computation, with over 15 years' experience working in applied research with a focus on medical interventions, and is an internationally recognised expert on ultrasound simulation.

- Formulates realistic yet tractable mathematical models, and deployable, tested and documented simulations. Excellent coding and software development skills developed through clinical/commercial deployment, satisfying ISO 13485/IEC 62304 standards.
- Communication skills refined through working in multi-disciplinary, international teams; lecturing and teaching at universities; delivering invited presentations at international conferences. Experienced in leading and managing complex interdisciplinary projects, as well as working in a team.
- 19 peer-reviewed journal papers (over 250 citations), a book chapter, acquisition of grants (over 500,000€), 12 invited presentations, industrial supervisor to three PhD students, an MSc student, and maintainer of widely used open source code, k-wave-python

Professional Experience

Aug 2026 -

Usono

Eindhoven, Netherlands / Hybrid

Senior ML Engineer — AI/ML Group

- Implementation of deep learning for motion analysis from ultrasound images
- Synthetic data generation from simulation data for machine learning models

Nov 2019 - May 2026

Fraunhofer-Institut für Digitale Medizin MEVIS

Bremen, Germany

Senior Research Scientist — Modelling & Simulation Group

- Develops large-scale simulations for microwave and ultrasound ablative therapies by implementing of high-performance numerical methods, enabling clinically relevant treatment planning at scale.
- Engineers fast ultrasound beamforming algorithms and a transcranial acoustic/elastic propagation simulator by leveraging GPU-accelerated signal-processing techniques, delivering real-time imaging performance and improved diagnostic accuracy.

Jun 2014 - Nov 2019

National Physical Laboratory

Teddington, United Kingdom

Senior Research Scientist — Ultrasound & Underwater Acoustics Group

- Established measurement-based simulation for nonlinear propagation through complex media by integrating empirical measurements into high-fidelity nonlinear computational models, enabling accurate predictions and incorporation into the IEC standard 63587.

Jun 2011 - Jun 2014

Institute of Cancer Research/The Royal Marsden Hospital

Sutton, United Kingdom

Post-Doctoral Research Associate — Therapeutic Ultrasound Group/Joint Department of Physics

- Developed an ultrasound-guided high-intensity focused ultrasound treatment planning system with a multi-element phased-array by integrating real-time ultrasonic imaging and adaptive beamforming control, enabling precise focus steering, dynamic treatment adjustments, and enhanced therapy safety.
- Implemented motion-management strategies for therapeutic ultrasound.

Jun 2008 - Jun 2011

University College London

London, United Kingdom

Post-Doctoral Research Associate — Ultrasound Group/Department of Mechanical Engineering

- Investigated the influence of cavitation on therapeutic ultrasound by employing numerical simulations and analytical modelling, enabling accurate prediction of cavitation thresholds and optimization of treatment efficacy and safety.

Education

2004–2008

PhD - Dynamical Systems

University College London, United Kingdom

2003–2004

MSc - Modern Applications of Mathematics

University of Bath, United Kingdom

2000–2003

BSc - Mathematics with Applied Math./Math. Physics

Imperial College London, United Kingdom

Professional Recognition

2020	IEEE IUS Challenge on Ultrasound Beamforming with Deep Learning (CUBDL) Joint first place in international machine learning challenge applied to ultrasound image reconstruction
2015-	International Expert • Member IEC/BSI Technical Committee 87 (Ultrasonics), in an individual capacity, • Associate IMA (Institute of Mathematics and its Applications), • Full member IOP (Institute of Physics)
Various	Enhanced Scholarships Enhanced funding for MSc (2004), PhD (2008) and post-doctoral work (2014) from UK funding agency EPSRC

Skills

Programming: python, C++, Matlab, OpenCL	Libraries: ITK, VTK, Boost, Eigen
DevOps: git, svn, github, gitlab, google test, pytest, make, cmake, visual studio	Computation: FEniCS, COMSOL
Languages: English (Native), German (B2.2) with permanent residency	